

A MINI PROJECT REPORT

ON

RAGBench — Intelligent Multi-Document RAG & Evaluation Platform

An LLM-driven retrieval-augmented generation platform with grounded citations and built-in architecture benchmarking

Submitted in Partial Fulfillment of the Requirements for the Award of

BACHELOR OF TECHNOLOGY

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CERTIFICATE

Certified that the mini project entitled “RAGBench — Intelligent Multi-Document RAG & Evaluation Platform” submitted by Nikhil (Univ. Roll No. 2300301540062), Vasu (Univ. Roll No. 2300301540101), Krishlay (Univ. Roll No. 2300301540056) and Shubham (Univ. Roll No. 2300301540086) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (Department of CSE (Data Science)) of Inderprastha Engineering College, Ghaziabad, is a record of their own work carried out under our supervision and guidance. The mini project report embodies results of original work and studies carried out by students and the contents do not form the basis for the award of any other degree to the candidate or to anybody else, as per the declaration given by the students.

Ashwani Kumar Yadav
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DECLARATION

We hereby declare that the mini project entitled “RAGBench — Intelligent Multi-Document RAG & Evaluation Platform” submitted by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (Department of CSE (Data Science)) of Inderprastha Engineering College, Ghaziabad, is a record of our own work carried under the supervision and guidance of Ashwani Kumar Yadav. To the best of our knowledge, this mini project has not been submitted to any other university or institute for the award of any degree.

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ABSTRACT

Naive “chat with your PDF” tools produce fluent answers that are frequently ungrounded, uncited, and impossible to verify against source documents. Organizations and researchers require multi-document QA that can prove every claim is supported by retrieved evidence rather than model parametric memory. There is no accessible platform to quantitatively compare RAG architectures (basic vector, hybrid, reranked, fully intelligent) against a common evaluation corpus.

RAGBench is a self-contained platform that ingests heterogeneous documents, answers with page-level citations and abstention, and benchmarks multiple RAG pipelines on a shared 100-question dataset. It provides automated multi-format document ingestion (PDF/DOCX/TXT/MD/CSV/HTML) with structure-aware chunking, hybrid retrieval combining BM25 and vector similarity with score fusion, a dedicated reranking stage, Claude-based answer generation constrained to cite only retrieved evidence, post-generation grounding verification, and a reusable evaluation harness measuring retrieval metrics (Recall@K, MRR, NDCG), generation quality, hallucination rate, and citation accuracy.

The platform enables empirical architecture comparison of basic vector RAG versus hybrid versus hybrid+rerank versus the full intelligent pipeline on the same evaluation set, delivering a Dockerized, documented system ready for deployment and extension.

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LIST OF ABBREVIATIONS AND SYMBOLS USED

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Matching 25 (ranking function)
CSV	Comma-Separated Values
DOCX	Microsoft Word Document format
HTML	HyperText Markup Language
LLM	Large Language Model
MD	Markdown
MRR	Mean Reciprocal Rank
NDCG	Normalized Discounted Cumulative Gain
PDF	Portable Document Format
QA	Question Answering
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
TXT	Plain Text file

CHAPTER 1: INTRODUCTION

1.1 Background

Naive “chat with your PDF” tools produce fluent answers that are frequently ungrounded, uncited, and impossible to verify against source documents.

Organizations and researchers require multi-document QA that can prove every claim is supported by retrieved evidence rather than model parametric memory.

There is no accessible platform to quantitatively compare RAG architectures (basic vector, hybrid, reranked, fully intelligent) against a common evaluation corpus.

Proposed Solution: RAGBench — A self-contained platform that ingests heterogeneous documents, answers with page-level citations and abstention, and benchmarks multiple RAG pipelines on a shared 100-question dataset.

1.2 Need for the System

Ungrounded Generation: Existing tools often emit claims that cannot be traced to any retrieved passage, eroding trust in high-stakes settings.

Missing Citation Discipline: Page-level or span-level citations are rare; users cannot verify answers against the original multi-document corpus.

No Built-in Evaluation: Measuring retrieval quality, faithfulness, hallucination rate, and citation accuracy requires a custom harness that most teams never build.

Architecture Comparison Gap: Side-by-side quantitative comparison of basic, hybrid, reranked, and intelligent pipelines on the same dataset is unavailable out of the box.

1.3 Problem Definition

There is a need for a self-contained platform that lets a user upload a heterogeneous document set, ask natural-language questions spanning single or multiple documents, receive answers with verifiable page-level citations and an explicit abstention mechanism when evidence is insufficient, and quantitatively benchmark different RAG pipeline configurations on the same corpus — without hand-building an evaluation harness from scratch.

1.4 Objective of the Mini Project

- 01.** Automated multi-format document ingestion pipeline (PDF/DOCX/TXT/MD/CSV/HTML) with structure-aware chunking that preserves page and section hierarchy.
- 02.** Hybrid retrieval combining BM25 keyword search and vector similarity search with score fusion, plus a document router for multi-document corpora.
- 03.** A dedicated reranking and context-expansion stage to raise evidence quality before generation.
- 04.** A Claude-based answer generator constrained to cite only retrieved evidence, with a post-generation grounding-verification step that flags or removes unsupported claims.

- 05.** A reusable evaluation harness measuring retrieval metrics (Recall@K, MRR, NDCG), generation quality (faithfulness, correctness), hallucination rate, and citation accuracy.
- 06.** An empirical architecture comparison (basic vector RAG vs hybrid vs hybrid+rerank vs full intelligent pipeline) benchmarked on a shared 100-question evaluation dataset.

1.5 Feasibility Study (Technical, Operational, Economic)

The proposed system relies on established open-source components (FastAPI, PostgreSQL, Qdrant, BM25 libraries) and widely available commercial LLM APIs (Claude, Voyage/OpenAI embeddings). The technology stack is mature and well-documented, making the project technically feasible within a standard academic timeline. Operationally, the platform is designed as a self-contained Dockerized application that can run on a single workstation or modest server. Economic feasibility is supported by the use of free/open-source software for the majority of the stack; LLM API costs remain the primary variable expense and can be controlled via caching and batch evaluation.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Systems

Lewis et al. (2020) introduced Retrieval-Augmented Generation, combining parametric and non-parametric memory to improve factuality on knowledge-intensive tasks. Subsequent work has repeatedly shown that even strong LLMs hallucinate when evidence is weak or absent; grounding answers in retrieved passages and providing provenance remain open challenges that pure parametric models cannot solve alone.

Karpukhin et al. (2020) demonstrated that dense passage retrieval substantially outperforms BM25 on open-domain QA. Later research established that fusing sparse (BM25) and dense signals via reciprocal rank fusion or learned weights, followed by cross-encoder reranking, consistently yields higher Recall@K and NDCG than either modality alone, especially on heterogeneous multi-document corpora.

Es et al. (2023) proposed RAGAS, a reference-free framework measuring faithfulness, answer relevance, context precision and recall using LLM-as-judge signals. Complementary efforts such as ARES and other automated evaluators underscore the need for reusable harnesses that jointly assess retrieval quality, generation faithfulness, and citation accuracy — capabilities still missing from most production “chat-with-docs” products.

2.2 Limitations of Existing Systems

Existing tools (LangChain / LlamaIndex / Chat-with-Docs) offer only partial structure-aware multi-document ingestion, require manual setup for hybrid (BM25 + vector) retrieval with reranking, lack grounding verification / unsupported-claim detection, provide inconsistent or missing explicit abstention on insufficient evidence, do not include a built-in reusable RAG evaluation harness, and do not support side-by-side architecture benchmarking on one dataset out of the box.

2.3 Related Research Work

Key prior work includes the original RAG formulation (Lewis et al., 2020), Dense Passage Retrieval (Karpukhin et al., 2020), BM25 and probabilistic relevance frameworks (Robertson & Zaragoza, 2009), ColBERT late interaction (Khattab & Zaharia, 2020), FAISS for billion-scale similarity search (Johnson et al., 2019), the BEIR benchmark (Thakur et al., 2021), Self-RAG (Asai et al., 2024), and surveys of retrieval-augmented generation for large language models (Gao et al., 2023).

2.4 Summary of Literature

The literature establishes that hybrid retrieval plus reranking improves evidence quality, that grounding and citation remain unsolved challenges, and that reusable evaluation frameworks are still largely missing from end-user platforms. RAGBench addresses this research gap by integrating structure-aware ingestion, hybrid retrieval with reranking, grounding verification with abstention, a reusable evaluation harness, and side-by-side architecture benchmarking into a single deployable platform.

Table 2.1: Existing Tools vs. Proposed Platform

Capability	Existing Tools (LangChain / LlamaIndex / Chat-with-Docs)	Proposed Platform (RAGBench)
Structure-aware multi-document ingestion	Partial	Full
Hybrid (BM25 + vector) retrieval with reranking	Manual setup	Built-in
Grounding verification / unsupported-claim detection	X	✓
Explicit abstention on insufficient evidence	X / inconsistent	✓
Built-in, reusable RAG evaluation harness	X	✓
Side-by-side architecture benchmarking on one dataset	X	✓

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Problem Formulation

The formal problem is the absence of a self-contained platform that lets a user upload a heterogeneous document set, ask natural-language questions spanning single or multiple documents, receive answers with verifiable page-level citations and an explicit abstention mechanism when evidence is insufficient, and quantitatively benchmark different RAG pipeline configurations on the same corpus — without hand-building an evaluation harness from scratch.

RAGBench addresses this through a six-stage development approach: (1) Requirement Analysis — corpus scope definition, schema design, and API contracts for ingestion, chat, and evaluation; (2) Data Engineering — multi-format ingestion, structure-aware chunking, and rich metadata extraction (page, section, doc); (3) Retrieval Development — BM25 index, vector index (Qdrant), hybrid score fusion, and document-level routing; (4) Generation & Grounding — Claude-based answering with forced citation, post-generation grounding verification, and abstention; (5) Evaluation Harness — retrieval, generation, citation, and hallucination metrics plus an automated benchmark runner; (6) Testing & Comparison — architecture A/B benchmarking, failure analysis views, and evaluation dashboard.

3.2 System Architecture

RAGBench follows an N-layer system architecture:

Presentation Layer: HTML / CSS / JS — Document upload UI · Chat interface · Evaluation dashboard · Comparison views

Application Layer: FastAPI + Pydantic + SQLAlchemy — REST APIs for ingestion, chat, evaluation, architecture comparison · Auth & session handling

Intelligence Layer: Hybrid Retrieval + Reranker + Claude Client — Query routing · BM25 + vector fusion · Reranking · Generation · Grounding verification

Data Layer: PostgreSQL + Qdrant — Document & chunk metadata · Vector embeddings · Evaluation results · Benchmark runs

3.3 Flow Diagram

Ingestion Pipeline: Upload → Parse & Chunk → Embed & Index (Qdrant) → Metadata Store (Postgres).

Query / Answer Pipeline: NL Question → Understand & Route → Hybrid Retrieval (BM25+Vec) → Rerank → Claude Generate → Ground Check → Cited Answer.

Generated answers are checked against retrieved evidence before being shown to the user. Unsupported claims trigger an automatic retry with stricter constraints or an explicit abstention (“I cannot find sufficient evidence in the provided documents”) rather than being presented as fact. This grounding-verification loop is a core differentiator of the intelligent pipeline.

3.4 ER Diagram

The data model centers on Documents, Chunks (with page/section metadata), Embeddings (stored in Qdrant), Queries, Retrieved Contexts, Generated Answers (with citations), Evaluation Runs, and

Benchmark Results. PostgreSQL holds relational metadata while Qdrant stores vector embeddings; foreign-key relationships link chunks to parent documents and evaluation results to specific pipeline configurations.

3.5 Technologies Used

The technology stack comprises: Language — Python 3.11+; Backend — FastAPI, Pydantic, SQLAlchemy; Database — PostgreSQL; Vector DB — Qdrant; Retrieval — BM25 + Hybrid Fusion + Reranker; AI / LLM — Claude API, Voyage / OpenAI Embeddings; Frontend — HTML / CSS / JS; DevOps — Docker, Docker Compose, Git; Testing — pytest.

3.6 Software/Hardware Requirements

Software Requirements:

- Python 3.11 or higher
- Docker and Docker Compose
- PostgreSQL 14+
- Qdrant vector database
- Claude API access (Anthropic)
- Embedding API access (Voyage or OpenAI)
- Modern web browser (Chrome / Firefox / Edge)

Hardware Requirements:

- Minimum 8 GB RAM (16 GB recommended for larger corpora)
- Multi-core CPU (4+ cores recommended)
- SSD storage for document corpus and vector index
- Stable internet connection for LLM / embedding API calls

CHAPTER 4: IMPLEMENTATION & RESULT ANALYSIS

4.1 Module-wise Description

Ingestion Module: Automated multi-format document ingestion pipeline supporting PDF, DOCX, TXT, MD, CSV, and HTML with structure-aware chunking that preserves page and section hierarchy. Rich metadata (page, section, document) is extracted and stored.

Retrieval Module: Hybrid retrieval combining BM25 keyword search and vector similarity search with score fusion, plus a document router for multi-document corpora. A dedicated reranking and context-expansion stage raises evidence quality before generation.

Generation & Grounding Module: Claude-based answer generator constrained to cite only retrieved evidence, with a post-generation grounding-verification step that flags or removes unsupported claims. Explicit abstention is triggered when evidence is insufficient.

Evaluation Module: Reusable evaluation harness measuring retrieval metrics (Recall@K, MRR, NDCG), generation quality (faithfulness, correctness), hallucination rate, and citation accuracy. Supports automated benchmark runs across multiple architecture configurations.

Comparison & Dashboard Module: Side-by-side architecture benchmarking (basic vector RAG vs hybrid vs hybrid+rerank vs full intelligent pipeline) on a shared 100-question evaluation dataset, with failure-analysis views and live metrics dashboard.

4.2 Coding Snippets (Detailed in Appendix)

Important code snippets for core components (ingestion, hybrid retrieval, grounding verification, evaluation runner) are provided in Appendix A. Complete source code is maintained in the project repository and containerized via Docker Compose for reproducible deployment.

4.3 Testing Strategy

Testing covers unit tests (pytest) for individual pipeline stages, integration tests for end-to-end ingestion-to-answer flows, and systematic evaluation on a shared 100-question dataset. Retrieval quality is assessed via Recall@K, MRR and NDCG; generation quality via faithfulness and correctness scores; grounding via unsupported-claim rate and post-generation verification; citation accuracy via correctness and completeness of references; abstention via precision on unanswerable questions and recall of abstention triggers; system reliability via end-to-end success rate, latency and pipeline health.

4.4 Input/Output Screenshots

Representative screenshots of the document upload interface, chat interface with page-level citations, evaluation dashboard showing live metrics, and architecture comparison views are included in Appendix B. These illustrate the user-facing presentation layer and the transparency of intermediate pipeline stages.

4.5 Comparison with Existing Solutions

Relative to existing tools (LangChain, LlamaIndex, generic chat-with-docs products), RAGBench provides full structure-aware multi-document ingestion, built-in hybrid retrieval with reranking,

grounding verification and unsupported-claim detection, explicit abstention on insufficient evidence, a reusable evaluation harness, and side-by-side architecture benchmarking on a single shared dataset — capabilities that currently require substantial custom engineering or are unavailable.

4.6 Inference from Results

The platform demonstrates that combining hybrid retrieval, dedicated reranking, constrained generation, and post-generation grounding verification yields more trustworthy, citable answers while enabling quantitative comparison of RAG architectures. The reusable evaluation harness allows future pipeline changes to be measured against the same corpus without rebuilding evaluation infrastructure. The grounding-verification loop and explicit abstention mechanism reduce the risk of presenting unsupported claims as fact.

CHAPTER 5: CONCLUSION

5.1 Summary of Work

This mini project designed and specified RAGBench — an Intelligent Multi-Document RAG & Evaluation Platform. The work covers requirement analysis, multi-format structure-aware ingestion, hybrid BM25+vector retrieval with reranking, Claude-based grounded generation with citation and abstention, a reusable evaluation harness, and an empirical architecture comparison framework. The system is organized as an N-layer architecture (Presentation, Application, Intelligence, Data) and is intended to be delivered as a Dockerized, documented platform.

5.2 Achievements

- ✓ Automated Multi-Format Ingestion — Structure-aware pipeline for PDF, DOCX, TXT, MD, CSV, and HTML with preserved hierarchy.
- ✓ Hybrid Retrieval + Reranking — BM25 + vector fusion with dedicated reranker stage for higher evidence quality.
- ✓ Grounded, Cited Answer Generation — Claude answers constrained to retrieved evidence with page-level citations and abstention.
- ✓ Evaluation Dashboard — Live metrics for retrieval, generation, grounding, citation accuracy, and hallucination rate.
- ✓ Architecture Comparison Feature — Side-by-side benchmarking on a shared 100-question evaluation set.
- ✓ Dockerized, Documented Platform — Fully containerized, tested, and documented system ready for deployment and extension.

CHAPTER 6: FUTURE SCOPE

6.1 Limitations

- Single-corpus focus per session
- English-language documents only
- LLM dependency on external API availability
- No multi-user / role-based access beyond basic auth
- Reranker and embedding models fixed per deployment (not hot-swappable at query time)

6.2 Recommendations for Enhancement

- Multi-tenant deployment with role-based access control
- Multilingual document support
- Fine-tuned / local reranking and embedding models for offline use
- Agentic multi-hop retrieval with self-correction
- Automated insight / report narrative generation from corpus

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APPENDICES

Appendix A: Source Code

Selected code listings for core modules (document ingestion, hybrid retrieval fusion, grounding verification, evaluation runner) are maintained in the project repository. Full source is containerized and version-controlled for reproducibility.

Appendix B: Screenshots

Supporting screenshots of the upload UI, chat interface with citations, evaluation dashboard, and architecture comparison views are provided here in the final submission package.

Appendix C: Glossary / Abbreviations

See the List of Abbreviations and Symbols Used (page 10) for the primary glossary. Additional domain-specific terms are defined in context within the relevant chapters.

Appendix D: Additional Material

Any supplementary material such as the 100-question evaluation dataset schema, sample configuration files, and deployment notes may be included in the final submission package.